



THE KING'S SCHOOL

2011 Higher School Certificate Trial Examination

Mathematics Extension 1

General Instructions

- Reading time – 5 minutes
- Working time – 2 hours
- Write using black or blue pen
- Board-approved calculators may be used
- A table of standard integrals is provided at the back of this paper
- All necessary working should be shown in every question

Total marks – 84

- Attempt Questions 1-7
- All questions are of equal value

Disclaimer

This is a Trial HSC Examination only. Whilst it reflects and mirrors both the format and topics of the HSC Examination designed by the NSW Board of Studies for the respective sections, there is no guarantee that the content of this exam exactly replicates the actual HSC Examination.

Examination Paper continues on the next page

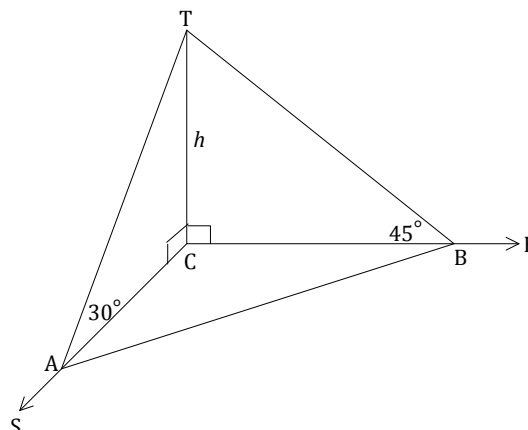
Total marks – 84
Attempt Questions 1-7
All questions are of equal value

Answer each question in a SEPARATE writing booklet. Extra writing booklets are available.

Question 1 (12 marks) Use a SEPARATE writing booklet.	Marks
(a) Find the derivative of $\tan(\ln x)$.	2
(b) A pair of parametric equations for a parabola is $x = 2t, y = -2t^2$ Find the Cartesian equation of the parabola.	2
(c) Find the smallest integer n for which $0.2^n < 12^{-2011}$	2
(d) Let $P(x) = x^3 - Ax^2 + Ax + 5$ (i) Find the remainder when $P(x)$ is divided by $x - 1$ (ii) Find the value of A if $x + 1$ is a factor of $P(x)$	1 2
(e) The equation $f(x) = \frac{x^3}{3} + x^2 - 5x - 1 = 0$ has a root near $x = 3$. Use one application of Newton's method to find an improved value for this root.	3

End of Question 1

(a)



A vertical tower CT of height h is due North of A and due West of B on horizontal ground ABC.

The elevations to the top T of the tower from A and B are 30° and 45° , respectively.

(i) Show that $AC = \sqrt{3} h$ 1

(ii) Find the bearing of B from A 2

(b) Sketch the graph of the function $y = -2 \sin^{-1} \left(\frac{x}{2} \right)$ clearly indicating the domain and range. 3

(c) Evaluate $\int_0^{\frac{\pi}{2}} 2 \sin^2 x - 1 + \sin x \, dx$ 3

(d) Solve the inequality $\frac{2x-1}{x+1} < -1$ 3

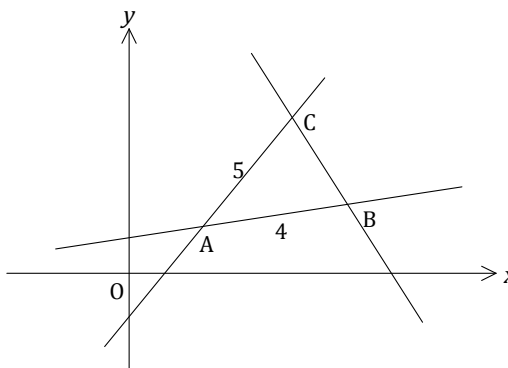
End of Question 2

(a) AC has equation $y = 3x - 2$

AB has equation $y = \frac{1}{2}x + 1$

Lines AC and AB meet line BC so that $AC = 5$ and $AB = 4$

Find the area of $\triangle ABC$



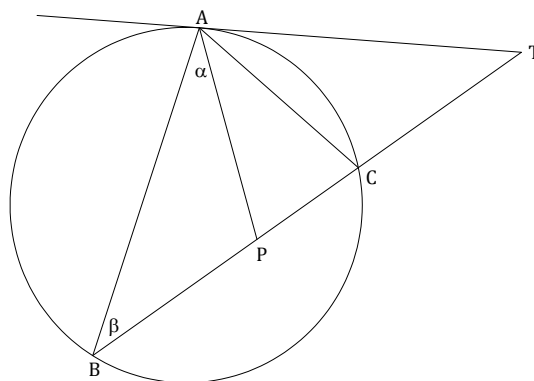
3

(b) AT is a tangent to the circle ABC

BCT is a straight line

P is the point on BCT so that PA bisects $\angle BAC$

Let $\angle BAP = \alpha$ and $\angle ABP = \beta$



(i) Explain why $\angle APC = \alpha + \beta$

1

(ii) Prove that $PT = AT$

3

(iii) Explain why $PT^2 = BT \times TC$

1

(c) Use the substitution $x = u^2$ to show that $\int_1^3 \frac{6}{(1+x)\sqrt{x}} dx = \pi$

4

End of Question 3

(a) Let $f(x) = \frac{2\sqrt{x}}{x+1}$

(i) State the domain of the function. **1**

(ii) Find any stationary points and determine their nature. **3**

(iii) Sketch the graph of $y = f(x)$

[YOU DO NOT NEED TO FIND POINTS OF INFLECTION] **2**

(b) $P(x) = x^3 + 4x^2 + Ax + 10 = 0$ has three real roots α, β, γ .

Two of these roots have a sum of 6.

Find the values of the roots. **3**

(c) The coefficient of x in the binomial expansion of $\left(x + \frac{a}{x^2}\right)^{10}$ is 15.

Find the value of a . **3**

End of Question 4

-
- (a) Prove by mathematical induction for positive integers n that
 $1 + 3 + 7 + \dots + (2^n - 1) = 2^{n+1} - (n + 2)$ **3**
- (b) A particle is moving on the x axis with its velocity v given by $v^2 = 2(8 - 2x - x^2)$
- (i) Prove that the motion is simple harmonic. **2**
- (ii) State the period of the motion. **1**
- (iii) Find the amplitude of the motion. **2**
- (c) Let $f(x) = -\ln(\sqrt{x} - 1)$
- (i) For what values of x is $f(x) > 0$? **2**
- (ii) Given that $f(x)$ decreases for all values of x in its domain, find explicitly the inverse function $y = f^{-1}(x)$ **2**

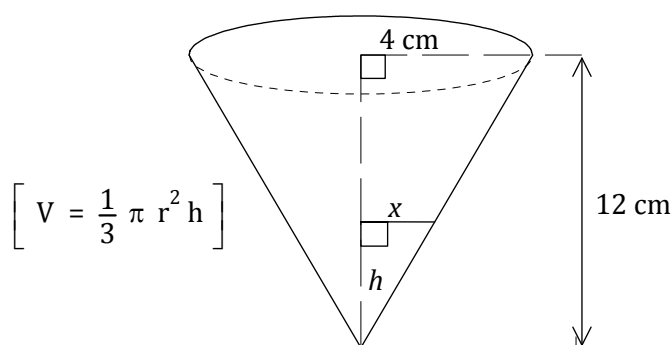
End of Question 5

- (a) Find the possible values of $\tan \theta$ if $9 \cos 2\theta + 7 \sin 2\theta = 11$ 2

- (b) The velocity, v m/s, for a particle moving on the x axis is given by $v = \frac{1}{6}(3 + 5e^{-2x})$
Initially the particle is at $x = 0$

Find the time taken for the particle to travel 1 metre. 3

(c)



A cone of radius 4 cm and height 12 cm is being filled with water at a constant rate of $2 \text{ cm}^3/\text{s}$.

Let the depth of the water in the cone after t s be h cm and the radius of the surface water be x cm.

- (i) Show that $x = \frac{h}{3}$ 1

- (ii) Find the rate at which the depth is increasing when the depth is 4 cm. 3

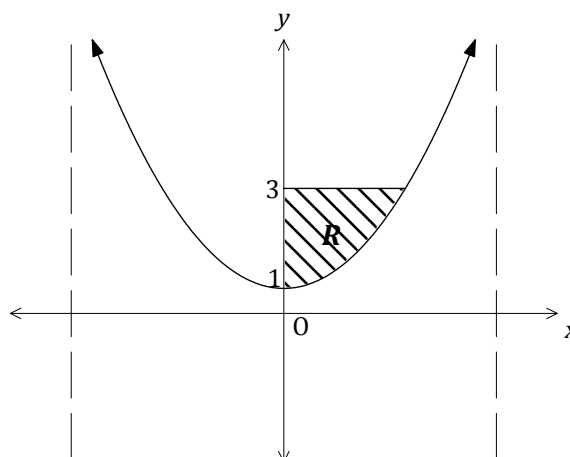
- (iii) Now suppose that the cone is being filled at a constant rate of $k \text{ cm}^3/\text{s}$ and also is leaking at the vertex of the cone at a variable rate of $\frac{\sqrt{h}}{10} \text{ cm}^3/\text{s}$.

It is observed that when the depth is 4 cm the depth is increasing at the rate of 0.036 cm/s . 3

Prove that the cone will eventually fill.

End of Question 6

(a)

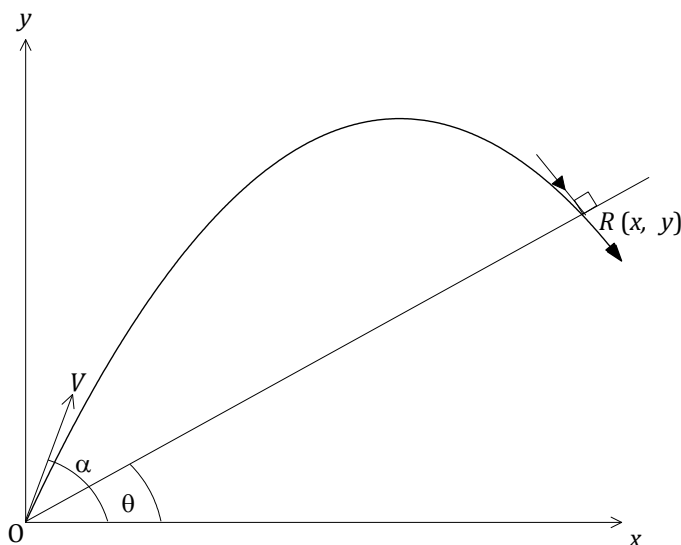


The sketch shows the shaded region **R** bounded by the curve $y = \frac{3}{\sqrt{9 - 2x^2}}$ and the y axis between $y = 1$ and $y = 3$.

- | | | |
|------|---|----------|
| (i) | Find the equations of the vertical asymptotes to the curve. | 1 |
| (ii) | Find the area of the region R . | 4 |

Question 7 continues on the next page

(b)



A particle is projected from O at an angle of elevation α with velocity V up an inclined plane which makes an angle θ with the horizontal.

The particle hits the plane at right angles at $R(x, y)$ at time T .

In usual notations, you may assume that

$$\dot{x} = V \cos \alpha$$

$$\dot{y} = -gt + V \sin \alpha$$

$$x = V \cos \alpha t$$

$$y = -g \frac{t^2}{2} + V \sin \alpha t$$

- | | |
|--|----------|
| (i) Show that at R , $y = x \tan \theta$ | 1 |
| (ii) Deduce that $gT = 2V(\sin \alpha - \cos \alpha \tan \theta)$ | 2 |
| (iii) Show that the horizontal component of velocity at $R(x, y)$ makes an angle θ with the inclined plane. | 1 |
| (iv) Prove that $\tan \alpha = \cot \theta + 2 \tan \theta$ | 3 |

End of Examination Paper

Standard Integrals

$$\int x^n dx = \frac{1}{n+1} x^{n+1}, \quad n \neq -1; \quad x \neq 0, \text{ if } n < 0$$

$$\int \frac{1}{x} dx = \ln x, \quad x > 0$$

$$\int e^{ax} dx = \frac{1}{a} e^{ax}, \quad a \neq 0$$

$$\int \cos ax dx = \frac{1}{a} \sin ax, \quad a \neq 0$$

$$\int \sin ax dx = -\frac{1}{a} \cos ax, \quad a \neq 0$$

$$\int \sec^2 ax dx = \frac{1}{a} \tan ax, \quad a \neq 0$$

$$\int \sec ax \tan ax dx = \frac{1}{a} \sec ax, \quad a \neq 0$$

$$\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \tan^{-1} \frac{x}{a}, \quad a \neq 0$$

$$\int \frac{1}{\sqrt{a^2 - x^2}} dx = \sin^{-1} \frac{x}{a}, \quad a > 0, \quad -a < x < a$$

$$\int \frac{1}{\sqrt{x^2 - a^2}} dx = \ln \left(x + \sqrt{x^2 - a^2} \right), \quad x > a > 0$$

$$\int \frac{1}{\sqrt{x^2 + a^2}} dx = \ln \left(x + \sqrt{x^2 + a^2} \right)$$

Note: $\ln x = \log_e x, \quad x > 0$

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Student Number



THE KING'S SCHOOL

2011 Higher School Certificate Trial Examination

Mathematics Extension 1

Question	Algebra and Number	Geometry	Functions	Trigonometry	Differential Calculus	Integral Calculus	Total
1	<i>c</i> 2		<i>b, d, e</i> 8		<i>a</i> 2		12
2	<i>d</i> 3		<i>b</i> 3	<i>a</i> 3		<i>c</i> 3	12
3		<i>b</i> 5	<i>a</i> 3			<i>c</i> 4	12
4			<i>a-i, iii, b, c</i> 9		<i>a-ii</i> 3		12
5	<i>a</i> 3		<i>c</i> 4		<i>b</i> 5		12
6				<i>a</i> 2	<i>b, c</i> 10		12
7			<i>a-i</i> 1		<i>b</i> 7	<i>a-ii</i> 4	12
Total	8	5	28	5	27	11	84

1) (a) $\frac{\sec^2(\ln x)}{x}$

(b) $x = \frac{x}{2} \Rightarrow y = -2\left(\frac{x}{2}\right)^2$ i.e. $y = -\frac{x^2}{2}$ or $x^2 = -2y$

(c) $\therefore \ln 0.2^n < \ln 12^{-2011}$

$\Rightarrow n \ln 0.2 < -2011 \ln 12$

$\therefore n > \frac{-2011 \ln 12}{\ln 0.2}$ since $\ln 0.2 < 0$

$= 3104.9 \dots$

$\therefore$ least n is 3105

(d) (i) $R = P(1) = 1 - A + A + S = 6$

(ii) $R = P(-1) = -1 - A - A + S = 0$

$\therefore 2A = 4$

$A = 2$

(e) $f'(x) = x^2 + 2x - 5$

$f(3) = 9 + 9 - 15 - 1 = 2$, $f'(3) = 9 + 6 - 5 = 10$

$\therefore x_1 = 3 - \frac{2}{10} = 2.8$

Question 2

(a) (i) In $\triangle ACT$, $\tan 30^\circ = \frac{h}{AC} = \frac{1}{\sqrt{3}}$

$$\therefore AC = h\sqrt{3}$$

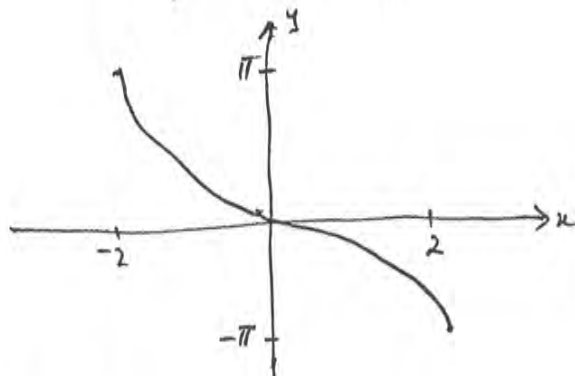
(ii) From $\triangle BCT$, $CB = h$

$$\therefore \text{In } \triangle ACB, \tan A = \frac{h}{h\sqrt{3}} = \frac{1}{\sqrt{3}}$$

$\therefore$ bearing of B from A is 30°

(b) $-1 \leq \frac{x}{2} \leq 1 \Rightarrow -2 \leq x \leq 2$

and range $-\pi \leq y \leq \pi$



$$\begin{aligned} \text{(c)} \quad I &= \int_0^{\pi/2} -\cos 2x + \sin x \, dx \\ &= \left[-\frac{\sin 2x}{2} - \cos x \right]_0^{\pi/2} = 0 - (0 - 1) = 1 \end{aligned}$$

(d) (LOTS OF ALTERNATIVES)

$$\begin{aligned} \text{If } x+1 > 0 & \quad \text{then } 2x-1 < -x-1 \\ \text{ie. } x > -1 & \quad \Rightarrow x < 0 \quad \therefore -1 < x < 0 \end{aligned}$$

& if $x < -1$ then $x > 0$ has no solution

$$\therefore -1 < x < 0$$

Question 3

(a) gradient $AC = 3$, gradient $AB = \frac{1}{2}$

$$\therefore \tan \angle CAB = \frac{3 - \frac{1}{2}}{1 + 3 \cdot \frac{1}{2}} = \frac{6-1}{2+3} = 1 \Rightarrow \angle CAB = \frac{\pi}{4}$$

$$\therefore \text{Area } \triangle ABC = \frac{1}{2} \cdot 5 \cdot 4 \cdot \sin \frac{\pi}{4} = \frac{10}{\sqrt{2}} \text{ will do} \\ = 5\sqrt{2}$$

(b) (i) $\angle APC = \angle BAP + \angle ABP$, ext $\angle$ thm in $\triangle ABP$
 $= \alpha + \beta$

[or, simply, ext $\angle$ thm in $\triangle ABP$]

(ii) $\angle CAP = \angle BAP = \alpha$, PA bisects $\angle BAC$
 $\angle TAC = \angle ABP = \beta$, alt seg thm

$$\therefore \text{From (i), in } \triangle APT, \angle P = \angle A = \alpha + \beta$$

$$\therefore \triangle APT \text{ is isosceles [base angles =]}$$

$$\Rightarrow PT = AT$$

(iii) Now $TA^2 = BT \times TC$, intersecting chord & tangent thm
 $= PT^2$ from (ii)

(c) $x = u^2$

$$x=1, u=1$$

$$\frac{dx}{du} = 2u$$

$$x=3, u=\sqrt{3}$$

$$\therefore I = \int_1^{\sqrt{3}} \frac{6 \cdot 2u}{u(1+u^2)} du$$

$$= 12 \int_1^{\sqrt{3}} \frac{1}{1+u^2} du$$

$$= 12 [\tan^{-1} u]_1^{\sqrt{3}}$$

$$= 12 \left(\frac{\pi}{3} - \frac{\pi}{4} \right) = \pi$$

Question 4

(a) (i) $x \geq 0$

$$(ii) f'(x) = \frac{(x+1) \cdot 2 \cdot \frac{1}{2} \cdot x^{-\frac{1}{2}} - 2\sqrt{x} \cdot 1}{(x+1)^2}$$

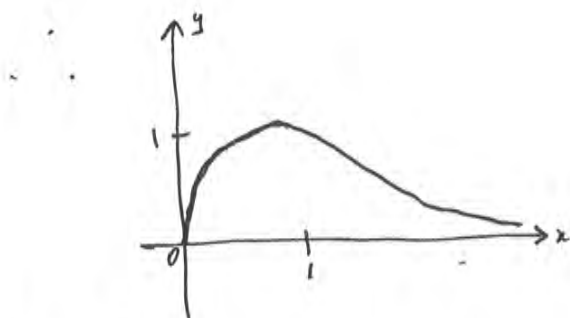
$$= \frac{x+1-2x}{\sqrt{x}(x+1)^2} = \frac{1-x}{\sqrt{x}(x+1)^2} = 0 \text{ if } x=1, f(1)=1$$

Now $f'(x) > 0$ if $x < 1$ and $f'(x) < 0$ if $x > 1 \Rightarrow$ ~~X~~

ie maximum turning point at $(1, 1)$

(iii) $f(0)=0$ & clearly $f(x) > 0$ for $x > 0$

$$\text{Further, } \lim_{x \rightarrow \infty} \frac{2\sqrt{x}}{x+1} = \lim_{x \rightarrow \infty} \frac{2\sqrt{x}}{x} = \lim_{x \rightarrow \infty} \frac{2}{\sqrt{x}} = 0$$



(b) $\alpha + \beta + \gamma = 6 + \gamma = -4 \therefore \gamma = -10$

product of roots $= \alpha\beta\gamma = -10 \therefore \alpha\beta = -10$

$\therefore \alpha + \beta = 6$ and $\alpha\beta = -10$

$\therefore \alpha(6-\alpha) = -10$ ie $\alpha^2 - 6\alpha + 10 = 0$

$$\therefore \alpha = \frac{6 \pm \sqrt{36-40}}{2} = \frac{6 \pm 4\sqrt{2}}{2} = 3 \pm 2\sqrt{2}$$

ie. roots are $-10, 3 \pm 2\sqrt{2}$

(c) $u_{k+1} = \binom{10}{k} x^{10-k} \left(\frac{a}{x^2}\right)^k = \binom{10}{k} a^k x^{10-k-2k}$

$$= \binom{10}{k} a^k x^{10-3k} \Rightarrow \text{coefficient of } x \text{ occurs when } 10-3k=1 \text{ or } k=3$$

$$\therefore \binom{10}{3} a^3 = 15 \Rightarrow a^3 = \frac{15}{120} = \frac{1}{8}$$

$$\therefore a = \frac{1}{2}$$

Question 5

(a) For $n=1$, $LS = 2-1=1$, $RS = 4-3=1$

$\therefore$ Assume $1+3+7+\dots+(2^n-1) = 2^{n+1} - (n+2)$ for some integer $n \geq 1$

Then $1+3+7+\dots+(2^n-1) + (2^{n+1}-1)$

$$= 2^{n+1} - (n+2) + 2^{n+1} - 1 \quad \text{using the assumption}$$

$$= 2 \cdot 2^{n+1} - (n+3)$$

$$= 2^{n+2} - (n+3)$$

$\therefore$ by induction it's true.

(b) (i) $\frac{1}{2}v^2 = 8 - 2x - x^2$

$$\therefore \ddot{x} = \frac{d(\frac{1}{2}v^2)}{dx} = -2 - 2x = -2(x+1)$$

is of the form $-n^2(x-b)$ $\therefore$ SHM

(ii) $n^2 = 2$, $n = \sqrt{2}$ $\therefore$ period $= \frac{2\pi}{\sqrt{2}} = \sqrt{2}\pi$

(iii) $v=0 \Rightarrow x^2 + 2x - 8 = 0$

$$(x-2)(x+4) = 0$$

$$\text{i.e. } x = 2, -4$$

$$\therefore \text{amp} = \frac{1}{2} \cdot 6 = 3$$

(c) (i) We need $\ln(\sqrt{x}-1) < 0 \Rightarrow 0 < \sqrt{x}-1 < 1$

$$\text{or } 1 < \sqrt{x} < 2$$

$$\therefore \text{for } 1 < x < 4$$

(ii) $y = f^{-1}(x)$: $x = -\ln(\sqrt{y}-1)$

$$\text{i.e. } \ln(\sqrt{y}-1) = -x$$

$$\therefore \sqrt{y}-1 = e^{-x}$$

$$\sqrt{y} = 1 + e^{-x}$$

$$\text{i.e. } f^{-1}(x) : y = (1 + e^{-x})^2$$

Question 6

(a) For ease, put $\tan \theta = t$

$$\tan \frac{9(1-t^2)}{1+t^2} + 7 \cdot \frac{2t}{1+t^2} = 11$$

$$\therefore 9 - 9t^2 + 14t = 11 + 11t^2$$

$$\text{or } 20t^2 - 14t + 2 = 0$$

$$\text{or } 10t^2 - 7t + 1 = 0$$

$$\therefore (5t - 1)(2t - 1) = 0$$

$$\therefore \tan \theta = \frac{1}{5} \text{ or } \frac{1}{2}$$

(b) $\frac{dx}{dt} = \frac{3 + 5e^{-2x}}{6}$

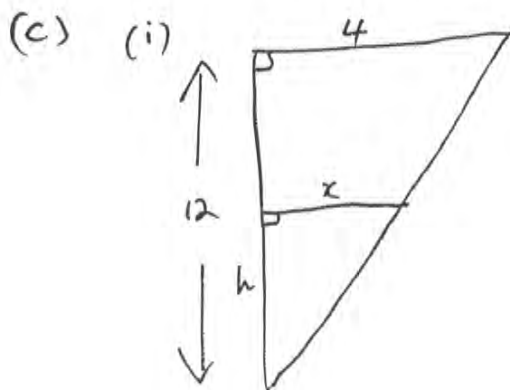
$$\therefore \frac{dt}{dx} = \frac{6}{3 + 5e^{-2x}} = \frac{6e^{2x}}{3e^{2x} + 5}$$

$$\therefore t = \int_0^1 \frac{6e^{2x}}{3e^{2x} + 5} dx \quad * \text{ Note } v > 0 \quad \forall x, t$$

$$= \left[\ln(3e^{2x} + 5) \right]_0^1$$

$$= \ln(3e^2 + 5) - \ln 8 \quad \text{will do}$$

$$= \ln\left(\frac{3e^2 + 5}{8}\right)$$



From similar Δs ,

$$\frac{x}{4} = \frac{h}{12} \Rightarrow x = \frac{h}{3}$$

$$(ii) \quad V = \frac{1}{3} \pi r^2 h \Rightarrow V = \frac{1}{3} \pi \left(\frac{h}{3}\right)^2 h = \frac{\pi h^3}{27}$$

$$\therefore \frac{dV}{dh} = \frac{\pi h^2}{9}$$

$$\text{Now } \frac{dh}{dt} = \frac{dh}{dV} \cdot \frac{dV}{dt}$$

$$= \frac{9}{\pi h^2} \cdot 2 = \frac{18}{\pi \times 4^2} \quad \text{when } h=4$$

$$= \frac{9}{8\pi} \text{ cm/s}$$

$$(iii) \quad \text{Now } \frac{dV}{dt} = k - \frac{\sqrt{h}}{10}$$

$$\text{and } \therefore \frac{dh}{dt} = \frac{9}{\pi h^2} \left(k - \frac{\sqrt{h}}{10} \right)$$

$$\therefore 0.036 = \frac{9}{\pi \times 4^2} \left(k - \frac{2}{10} \right)$$

$$\Rightarrow k = 16\pi \times 0.004 + 0.2 = 0.401 - \dots$$

$$\therefore \frac{dh}{dt} = \frac{9}{\pi h^2} \left(0.4 - \frac{\sqrt{h}}{10} \right) \quad \text{very nearly}$$

$$\text{But } \max h = 12 \quad \& \quad \frac{\sqrt{12}}{10} = 0.346 - \dots < 0.4$$

$$\therefore \text{ for all values of } 0 < h \leq 12, \quad \frac{dh}{dt} > 0$$

$\therefore$ cone will eventually fill

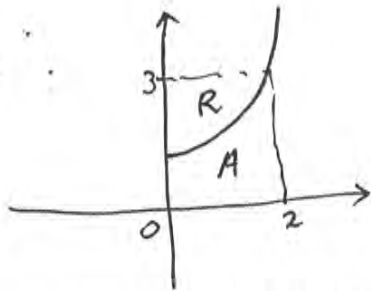
Question 7

(a) (i) Vertical asymptotes occur when $9 - 2x^2 = 0$

$$\text{i.e. } 2x^2 = 9$$

$$\therefore \text{asymptotes are } x = \pm \frac{3}{\sqrt{2}}$$

(ii) When $y = 3$, $9 - 2x^2 = 1$
or $2x^2 = 8$, $x^2 = 4 \Rightarrow x = 2$ for R



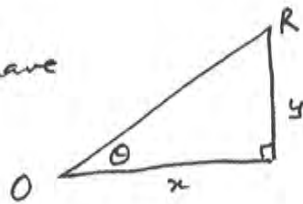
$$\therefore \text{area } R = 3 \times 2 - A$$

$$\Rightarrow \text{area } R = 6 - \int_0^2 \frac{3}{\sqrt{9-2x^2}} dx$$

$$= 6 - \frac{3}{\sqrt{2}} \left[\sin^{-1} \frac{\sqrt{2}x}{3} \right]_0^2$$

$$= 6 - \frac{3}{\sqrt{2}} \sin^{-1} \left(\frac{2\sqrt{2}}{3} \right)$$

(b) (i) at R we have



$$\therefore \tan \theta = \frac{y}{x}$$

$$\text{i.e. } y = x \tan \theta$$

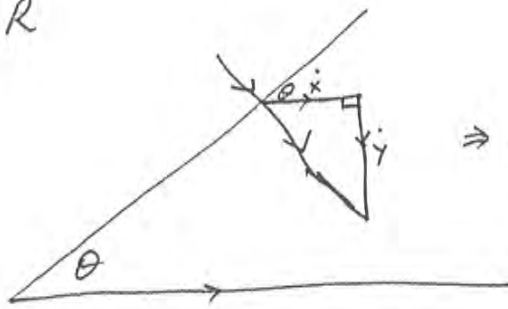
(ii) at R, $x = V \cos \theta$ and $y = -g \frac{T^2}{2} + V \sin \theta = x \tan \theta$

$$\therefore -g \frac{T^2}{2} + V \sin \theta = V \cos \theta \tan \theta$$

$$\Rightarrow gT = 2V \sin \theta - 2V \cos \theta \tan \theta$$

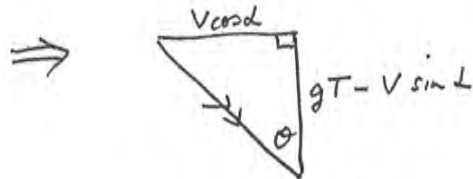
$$\text{i.e. } gT = 2V (\sin \theta - \cos \theta \tan \theta)$$

(iii) at R



$\Rightarrow$ angle between plane & x is θ ,
corresponding $\angle$ s in $\parallel$ lines
[lots of others]

(iv) at R, $\dot{x} = V \cos \alpha$, $\dot{y} = -gT + V \sin \alpha < 0$



$$\therefore \tan \theta = \frac{V \cos \alpha}{gT - V \sin \alpha}$$

$$\therefore gT - V \sin \alpha = \frac{V \cos \alpha}{\tan \theta} = V \cos \alpha \cot \theta$$

$$\therefore \text{from (ii), } V \cos \alpha \cot \theta = V \sin \alpha - 2V \cos \alpha \tan \theta$$

$$\therefore \cot \theta = \tan \alpha - 2 \tan \theta, \text{ dividing by } \cos \alpha$$

$$\text{i.e. } \tan \alpha = \cot \theta + 2 \tan \theta$$